Lab 4.1 – Inspection Tools

At the end of this lab exercise, Participants will be able to:

1. Determine if the pins are in the connector using a Brightness Tool

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

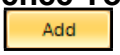
2. Easy Builder User Interface Application Step 2
    1. Inspect Part
    2. Presence / Absence Tools – Brightness

Follow the steps below to complete the lab exercise:

1. Load the **MyLocator** job from the previous lab.

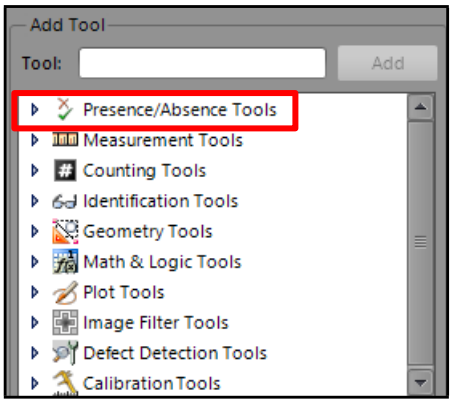
   NOTE: *Confirm that your demo plate is on the PASS side and that you are not in Live mode.*

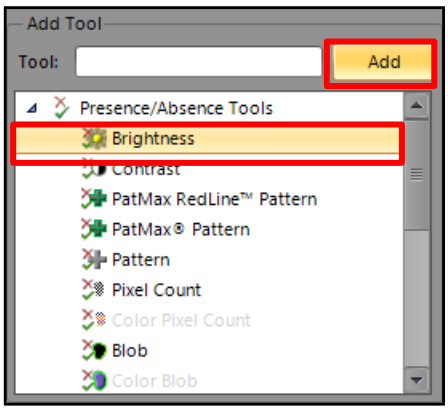
2. Save the job as **MyBrightness** in the folder you created on the desktop in Lab #2.
3. Click the **Inspect Part** 

Application Step button.
4. Expand the **Presence/Absence Tools** in the Inspection Tools and select **Brightness**. Click the **Add** 

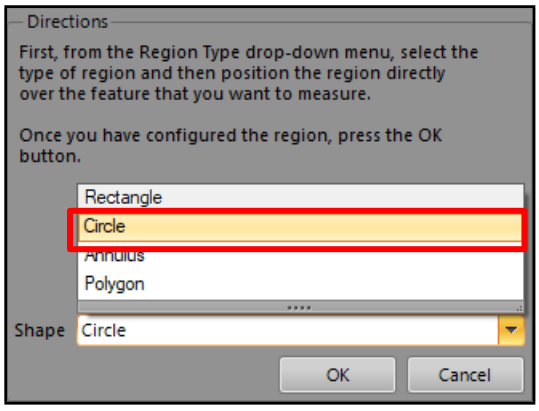
button.

   NOTE: *You can also double click on the tool. This will also add it on.*





5. Select **Circle** as the Shape.



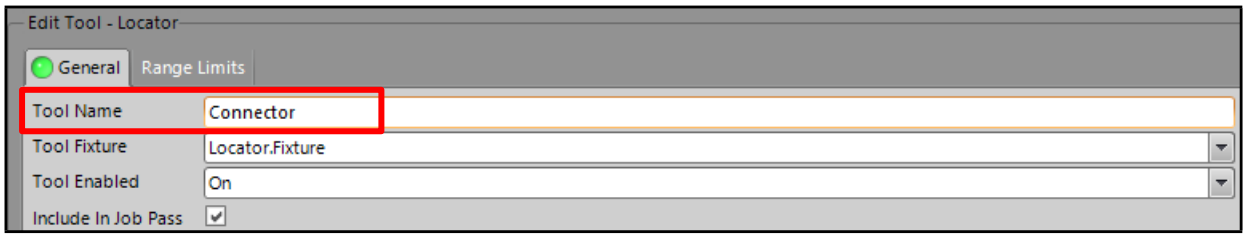
6. Move your circle over the Connector in the lower right corner. The Search area will be outlined in purple.
   NOTE: *Reshape the circle to include only the inside area where the pins will be – do not include any of the white from the side of the connector.*



7. Click the **OK** OK button.
   The Search area will be outlined in green.

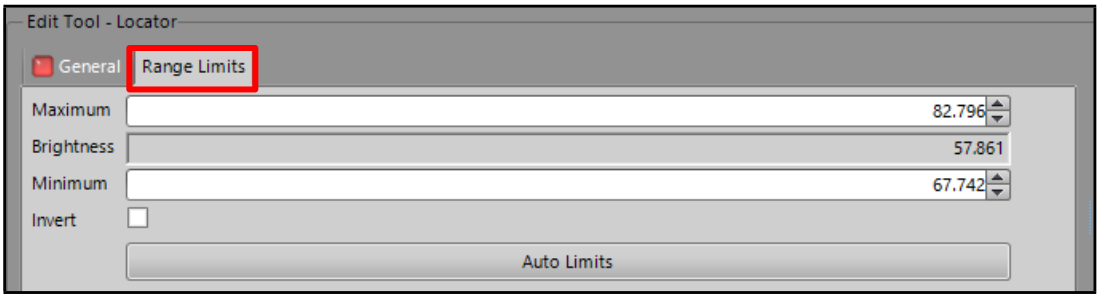


8. Change the Tool Name from Brightness_1 to **Connector**.
   NOTE: *To update the Palette Pane's Result tab, press the **<Enter>** key or click outside the box.*

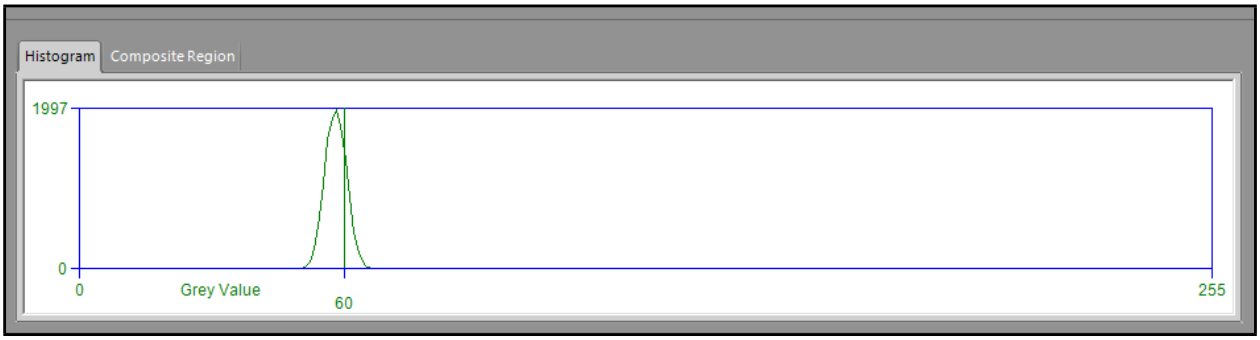


9. Click the **Range Limits** tab and test the range of Minimum-Maximum for Pass / Fail.

NOTE: *Change the demo plate over to the **Fail** side and acquire a new image.*



10. Confirm the Brightness value is outside the Minimum-Maximum values on the Fail side.

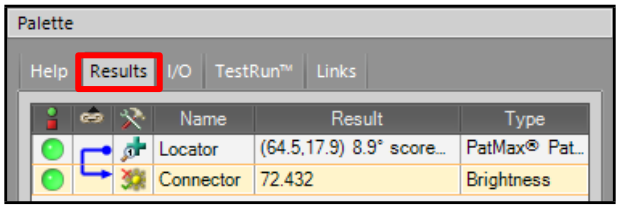


NOTE: *The LED graphic on the General tab will change from Green to Red when the image fails.*



11. Confirm the tool has been fixtured and that the graphic in the Palette pane on the Results tab is Chained to the Locator tool.
NOTE: *This is an automatic function.*





12. Save the job.

Lab 4.2 – Inspection Tools

At the end of this lab exercise, Participants will be able to:

1. Measure the distance of the camera body

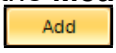
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

2. Easy Builder User Interface Application Step 2
    1. Inspect Part
    2. Measurement Tools – Distance

Follow the steps below to complete the lab exercise:

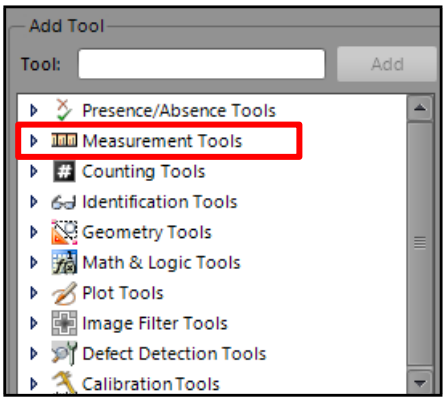
1. Continue with the **MyBrightness** job from the previous lab.
2. Save the job as **MyMeasure** in the folder that you created on the desktop in Lab #2.

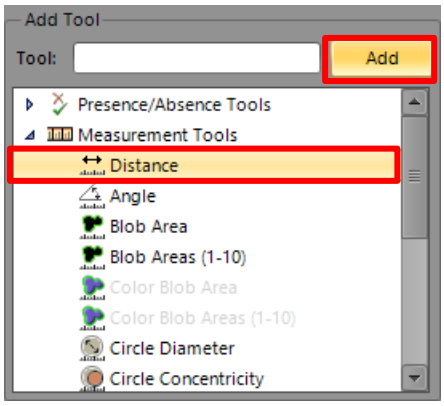
   NOTE: *Confirm that your demo plate is on the PASS side and that you are not in Live mode.*
3. Continue the Inspect Part 

Application Step.
4. Expand the **Measurement Tools** in the Inspection Tools and select **Distance**. Click the **Add** 

button.

   NOTE: *You can also double click on the tool. This will also add it on.*



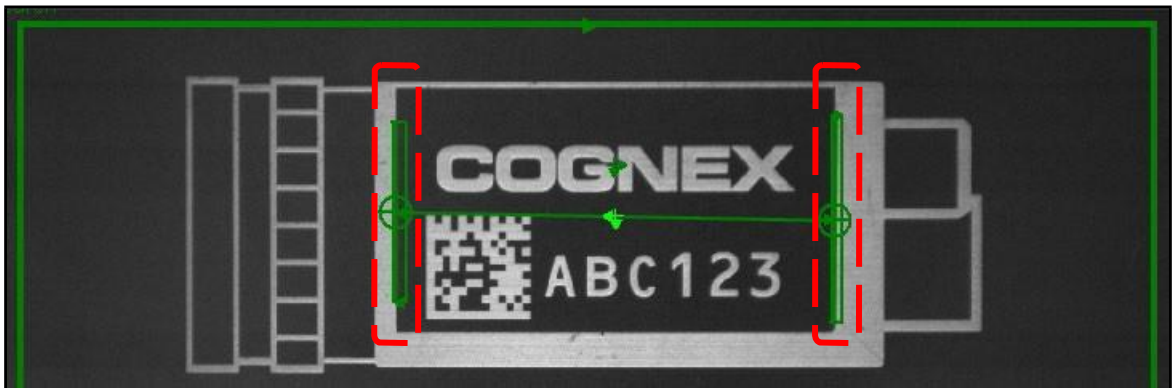


Smart Features (Blue Edges) will appear in the Image view window.



5. Click on the COGNEX cover's **Left** and **Right Blue Edges** (Smart Features).

   NOTE: *Once the second edge is selected the tool will be automatically created on the part, and the edges will turn green.*



6. Confirm the graphic is correct in the Image view window.

   NOTE: *By selecting a different starting point on the Smart Features, you **may** have a graphic that looks different, but it will measure the same distance / width*.

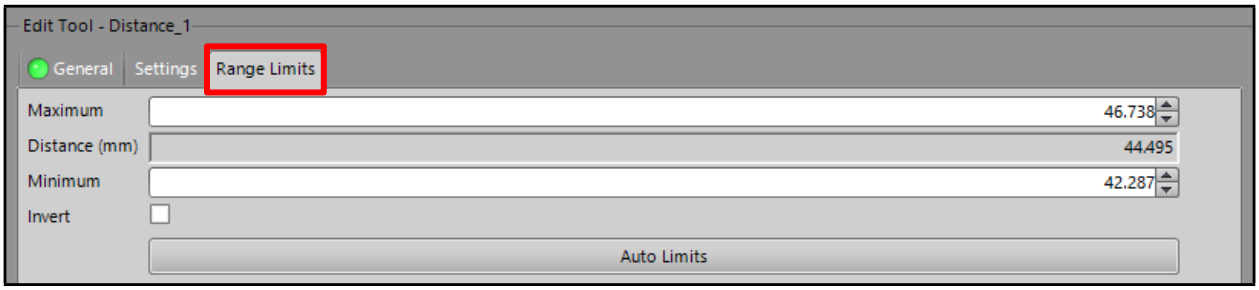
7. Change the Tool Name from Distance1 to **Width_1**.

   NOTE: *To update the Palette Pane's Result tab, press the **<Enter>** key or click outside the box*.



8. Click the **Range Limits** tab and test the range of Minimum-Maximum for Pass Fail.

   NOTE: *Turn the Camera demo plate over to the Fail side and acquire a new image. If a calibration was done the units are in **mm** not pixels.*



   CAUTION: *If one of the Edges tools reads Not Present in the Results Tab the Distance will output -0-.*
   SOLUTION: *Move or resize the tool to include the edge.*

9. Confirm the Distance value is outside the Minimum-Maximum values on the Fail side.

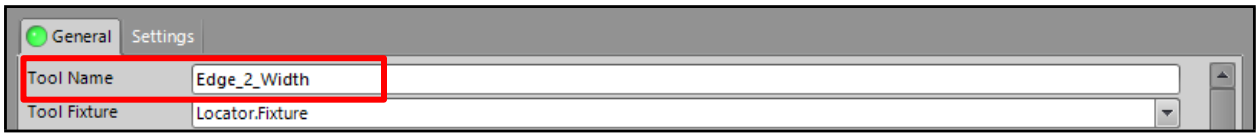
NOTE: *The LED graphic on the General tab will change from Green to Red when the image fails.*



10. Change the Tool Name from Edge_1 to **Edge_1_Width**.

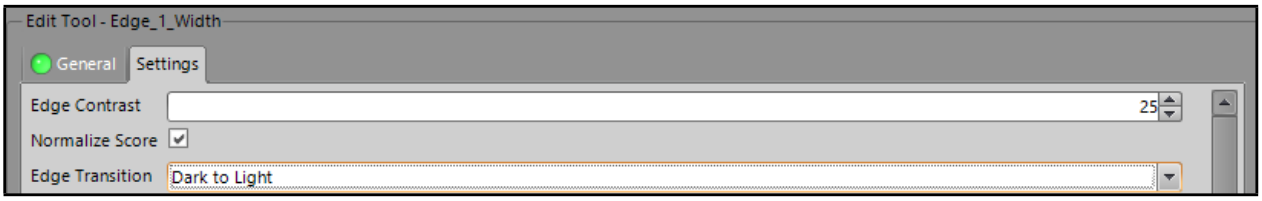


11. Rename Edge_2 to **Edge_2_Width**.

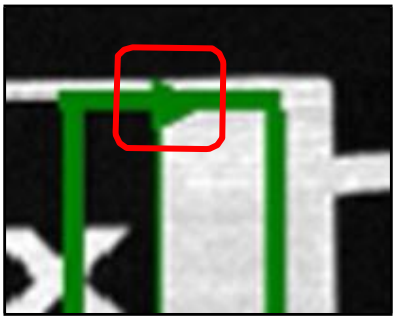


12. Click the **Settings** tab for both the Edge_1_Width and the Edge_2_Width tool and change the **Edge Transition** from *Both* to *Dark to Light*.



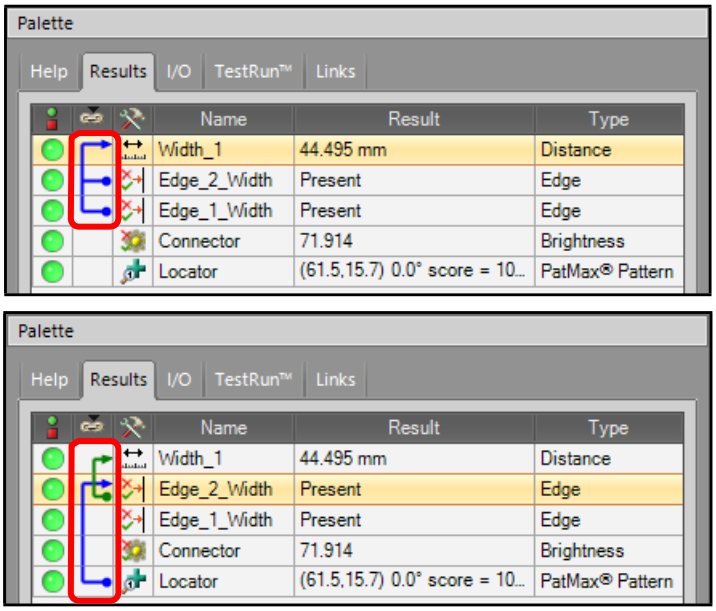


NOTE: *The Arrow indicates the scan direction for Edge Transition.*



13. Confirm the tool has been fixtured and that the graphic in the Palette pane is Chained to the Locator tool.

NOTE: *The Width_1 tool is **Referenced** to the Edge tools and the Edge tools are each **Fixtured** to the Locator tool. Click on each tool to see the different chain.*



14. Save the job.

Lab 4.3 – Inspection Tools

At the end of this lab exercise, Participants will be able to:

1. Determine the number of LEDs on the back of the camera

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

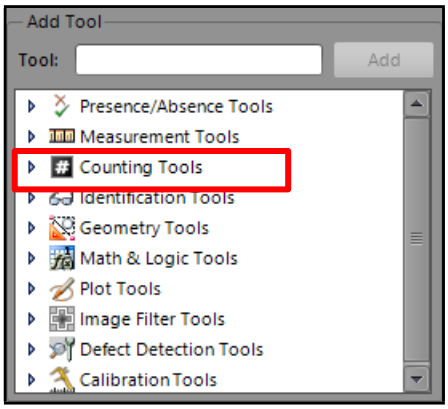
2. Easy Builder User Interface Application Step 2
    1. Inspect Part
    2. Counting Tools – Blobs

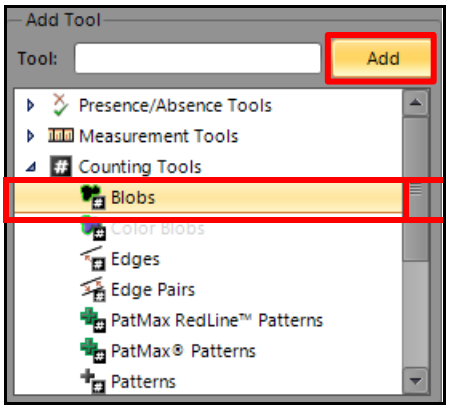
Follow the steps below to complete the lab exercise:

1. Continue with the **MyMeasure** job from the previous lab.
   NOTE: *Confirm that your demo plate is on the PASS side and that you are not in Live mode.*
2. Save the job as **MyCount** in the folder created in Lab #2.
3. Continue the **Inspect Part** 

Application Step.
4. Expand the **Counting Tools** in the Inspection Tools and select **Blobs**. Click the **Add** 

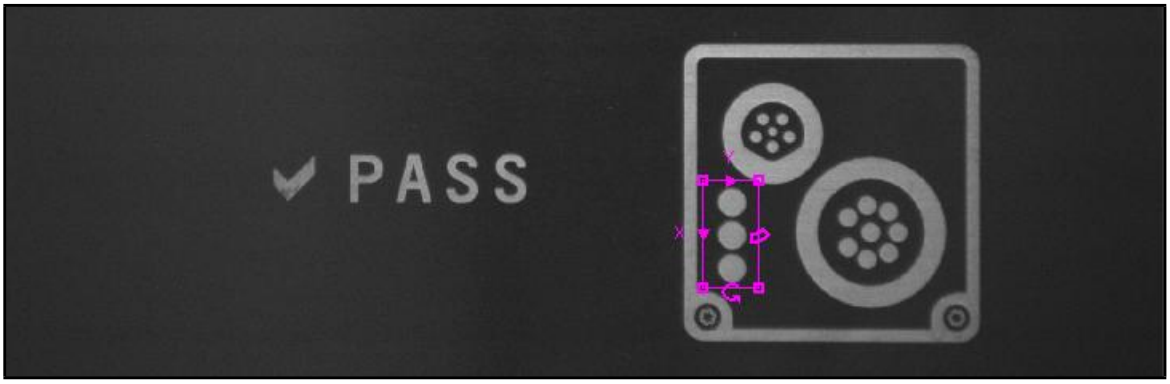
button.
   NOTE: *You can also double click on the tool. This will also add it on.*





5. Place the Rectangle tool around the LED features to be inspected and click the **OK** 

button.

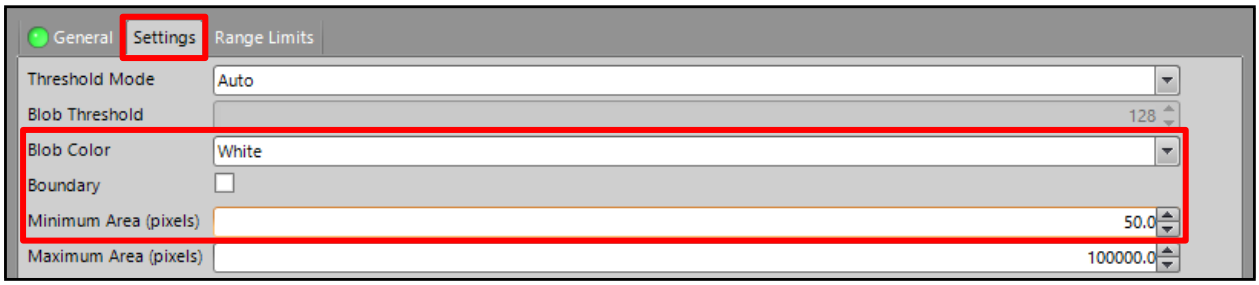


6. Change the Tool Name from Blobs_1 to **Count_LEDs.**

   NOTE: *This tool should work 'out of the box'. Next, you will set some parameters to make it even more robust.*

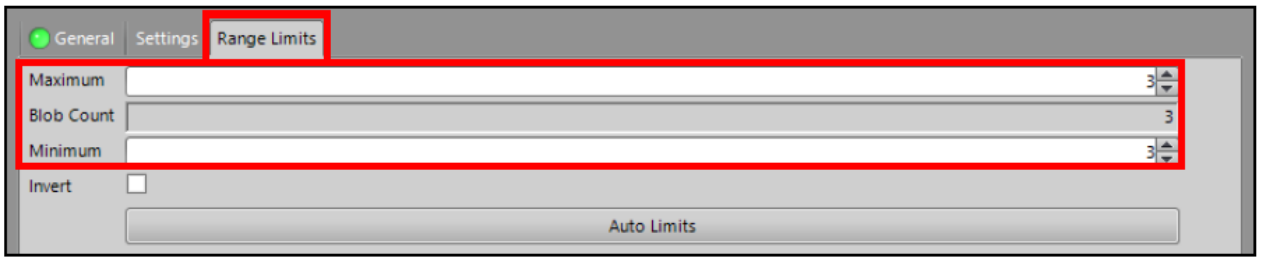


7. Set the following parameters on the **Settings** tab:
   - **Blob Color** = *White* (looks for light LED features)
   - **Boundary** = *unchecked*
   - **Minimum Area (pixels)** = *50* (counts the small LEDs; adjust it as needed)



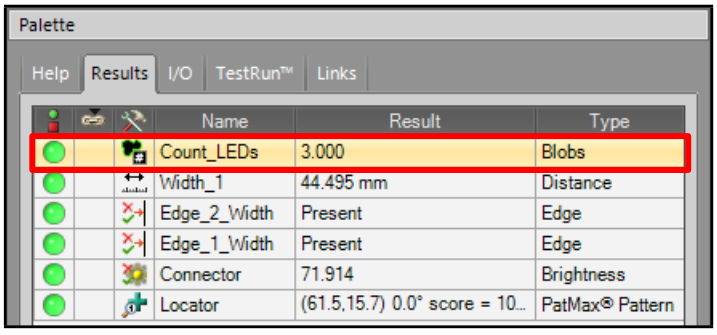
8. Click the **Range Limits** tab and set the **Minimum** and **Maximum** tolerances to *3***.**

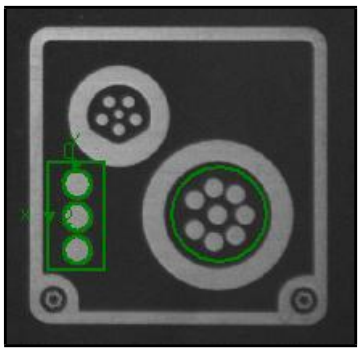
   NOTE: *These settings mean that the tool will only pass if there are exactly three LED features present inside the tool.*



9. On the **Results** tab in the Palette, click the **Count_LEDs** entry and confirm that the tool is passing with a result of **3**.

Passing

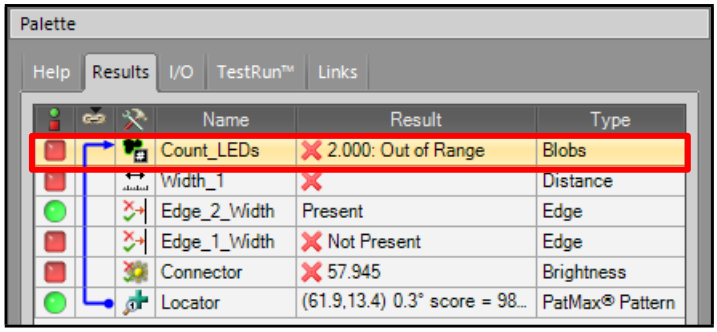


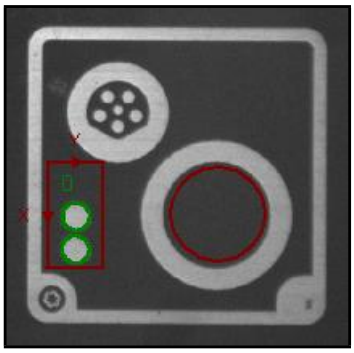


10. Turn the plate over to the FAIL side and acquire a new image. Confirm the Count_LEDs tool is failing with a result of **2: Out of Range**.

    NOTE: *The tool fails because there are only two LED features found instead of the expected three.*

    Failing

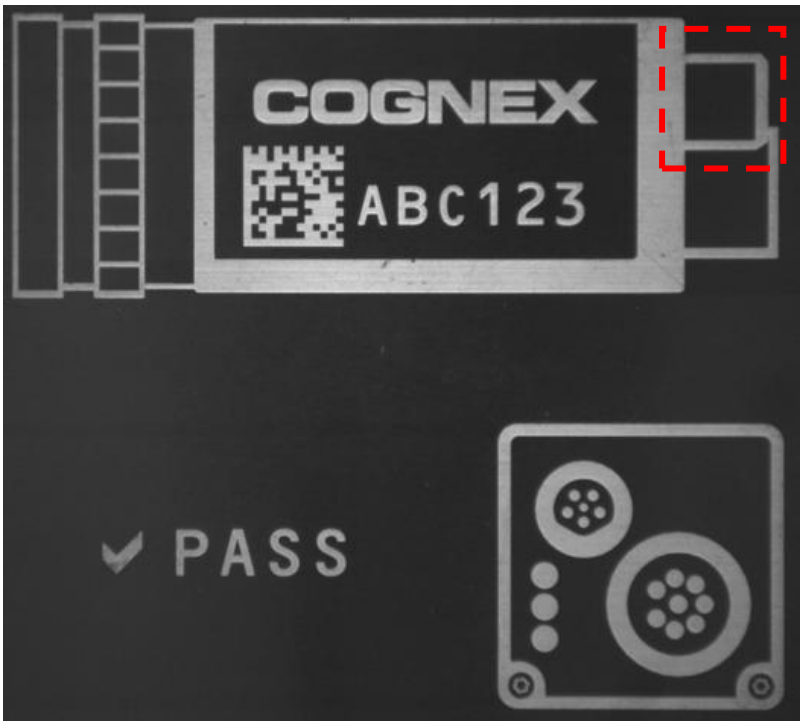




11. Save the job.

Lab 4.4 – Exercises

1. **Continue with previous job - MyCount**, and save it as **MyLab-4_YourName**, can you use a proper tool to inspect the area highlighted in red?



2. **Take a screenshot of the image view and the Palette. Submit the job and the screenshot on Blackboard.**